

L.J Institute of Engineering and Technology, Ahmedabad. CN Practice book (SEM-V-2025 CE,IT,CSD,AIML,AIDS,CS&IT,CSE,CST,CEA,RAI Engineering) Note : This Practic book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.								
Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
Unit-1 Introduction								
1	1	The first Network was called	D	1	CNNET	NSFNET	ASAPNET	ARPANET
2	1	Three or more devices share a link in _____ connection.	B	1	Unipoint	Multipoint	Point to point	Simplex
3	1	Two devices are in network if _____	A	1	a process in one device is able to exchange information with a process in another device	a process is running on both devices	PIDs of the processes running of different devices are same	a process is active and another is inactive
4	1	_____ can determine the category of a network.	D	1	Size	Ownership	The Physical Architecture	All of the above
5	1	In computer networks, LAN stands for _____ Area Network	A	1	Local	Live	Long	Length
6	1	Lower layer is responsible for _____ (on the receiver side)	B	1	The letter is picked up, removed from envelope, and read	The letter is delivered from the carrier to the post oddice	The letter is carried from the post office to the mailbox.	None of the above.
7	1	Which layer is not present in TCP/IP?	D	1	Application layer	Internet Layer	Transpot Layer	Presentation Layer
8	1	In computer networks, MAN stands for _____ Area Network.	C	1	Master	Maximum	Metropolitan	Mini
9	1	Which of the follwing are network services?	D	1	File Service	Print Service	Database Service	All of the above
10	1	TCP/IP has how many layers?	B	1	5	4	6	7
11	1	The _____ layer changes bits onto electromegnetic signals.	A	1	Physical	Transport	Data Link	Presentation
12	1	A device that is used to connect number of LANs	A	1	Router	Repeter	Bridge	All of the above
13	1	A gorup of computers and other devices connected together is called network, and concept of connected computers sharing resources is called	A	1	Networking	Inter-Network	Inter-Connection	Computer Group
14	1	Which is main function of Transport layer?	B	1	Node to Node delivery	End to End delivery	Synchronization	All of the above
15	1	Which of the following networking hardware can operate at anyone of the seven layers of the open systems interconnection model?	B	1	Router	Getway	Switch	Modem
16	1	Which of the following represents Latency?	A	1	propagation time +transmission time +queuing time + processing delay	propagation time -transmission time +queuing time + processing delay	propagation time +transmission time +queuing time - processing delay	propagation time +transmission time - queuing time - processing delay
17	1	What is the time needed by each device on a network, to hold the message before processing, called as?	B	1	Propagation Time	Queuing time	Transmission Tlme	Jitter
18	1	Which of the following devices can help to extend the range of a wireless router?	A	1	Wireless Repeter	Wireless Adapter	Wimax	Network Interface Card
19	1	What does ARPANET stand for?	B	1	Automated Research Projects Agency Network	Advanced Research Projects Agency Network	American Research Projects Agency Network	American Research Projects Automated Network
20	1	What does a set of rules define?	D	1	SMTP	FTP	IAMP	Protocol
21	1	Identify the total versions of IP.	B	1	1	2	4	6
22	1	Which of the following layer isn't present in the TCP/IP model but is included in the OSI model?	B	1	Network Layer	Session Layer	Application Layer	Transport Layer
23	1	What is the length of the IPv4 address?	C	1	4 Bits	6 Bits	32 Bits	16 Bits
24	1	Identify the device used to boost up a weak signal.	B	1	Modem	Repeater	Switch	Router
25	1	What is the length of the IPv6 address?	A	1	16 Byte	12 Byte	18 Byte	10 Byte
26	1	MAC address is of _____ and stored in _____	A	1	48 bits, NIC	48 bits, windows	48 bytes, Linux	48 bytes, NIC
27	1	What is the propagation time if the distance between the two points is 12000 km? Assume the propagation speed to be 2.4X10 ⁸ m/s in cable.	50 ms	2				
28	1	What is the Transmission time for a 2.5KB message if the bandwidth of the network is 1 Gbps?	0.02 ms	2				
29	1	How long does it take a packet of length 1,000 bytes to propagate over a link of distance 2,500 km, propagation speed 2.5 · 10 ⁸ m/s, and transmission rate 2 Mbps?	14 ms	2				
30	1	Suppose Host A wants to send a large file to Host B. The path from Host A to Host B has three links, of rates R1 = 500 kbps, R2 = 2 Mbps, and R3 = 1 Mbps. A) Assuming no other traffic in the network, what is the throughput for the file transfer? B) Suppose the file is 4 million bytes. Dividing the file size by the throughput, roughly how long will it take to transfer the file to Host B? C) Repeat (a) and (b), but now with R2 reduced to 100 kbps.		4				
31	1	Suppose Device A wants to send a large file to Device B. The path from Device A to Device B has three links, of rates R1 = 700 kbps, R2 = 5 Mbps, and R3 = 1 Mbps. A) Assuming no other traffic in the network, what is the throughput for the file transfer? B) Suppose the file is 14 million bytes. Dividing the file size by the throughput, roughly how long will it take to transfer the file to Device B? C) Repeat (a) and (b), but now with R2 reduced to 100 kbps		4				
32	1	We wish to send a message of size 150,000 bytes over the network. There are four hops, each of length 20km and running at 100Mb/s. Before sending we split the message into 1500 byte packets. What is the end-to-end delay of the message?		4				
33	1	Two hosts are connected via a packet switch with 10 ⁷ bits per second links. Each link has a propagation delay of 20 micro-seconds. The switch begins forwarding a packet 35 microseconds after it receives the same. If 10000 bits of data are to be transmitted between the two hosts using a packet size of 5000 bits,what is the time elapsed between the transmission of the first bit of data and the reception of the last bit of the data in microseconds?		3				
34	1	Two hosts are connected via a packet switch with 10 ⁷ bits per second links. Each link has a propagation delay of 50 micro-seconds. The switch begins forwarding a packet 45 microseconds after it receives the same. If 3000 bits of data are to be transmitted between the two hosts, what is the time elapsed between the transmission of the first bit of data and the reception of the last bit of the data in microseconds?		4				
35	1	Suppose Five devices A,B,C,D and E are connected in mesh topology. A wants to send data to D via C. how much time it would be required if A is sending 4000 bytes packet and length of each link is 1000 km and propagation speed 2.5*10 ⁸ m/s, and transmission rate 4 Mbps? Consider the processing and queing time faced by C is zero.		3				
36	1	Suppose Five devices D1, D2, D3, D4 and D5 are connected in a mesh topology. D1 wants to send the data to D5 via D2 and then D4. how much time would it be required in milliseconds, if D1 is sending 4000 bytes packet and length of each link is 10000 km and propagation speed 2.5*10 ⁸ m/s, and transmission rate 2 Mbps?		3				

L.J Institute of Engineering and Technology, Ahmedabad. CN Practice book (SEM-V-2025 CE,IT,CSD,AIML,AIDS,CS&IT,CSE,CST,CEA,RAI Engineering)								
Note : This Practic book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.								
Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
37	1	Suppose Host A wants to send a large file of 40,000 bytes to Host B. The path from Host A to Host B has three links , of rates R1 = 500 kbps, R2 = 1 Mbps, and R3 = 2 Mbps. A) Assuming no other traffic in the network, what is the throughput for the file transfer? B) Suppose A and B are connected to each other via Link R2. Now A wants to send packets of 10,000 bytes one by one to send entire file. when B receives all the packets it sends acknowledgement of 20bytes to A. Find total delay. Propagation delay is 20ms for each link.		3				
38	1	Consider a source computer (S) transmitting a file of size 10^6 bits to a destination computer (D) over a network of two routers (R1 and R2) and three links (L1, L2, and L3). L1 connects S to R1; L2 connects R1 to R2; and L3 connects R2 to D. Let each link be of length 1000 km. Assume signals travel over each link at a speed of 2×10^8 meters per second. Assume that the link bandwidth on each link is 100 Kbps. Let the file be broken down into 1000 packets each of size 1000 bits. Find the total sum of transmission and propagation delays in transmitting the file from S to D in milliseconds.		4				
39	1	We wish to send a message of size 150,000 bytes over the network. There are four hops, each of length 20km and running at 100Mb/s. Before sending we split the message into 1500 byte packets. What is the end-to-end delay of the message? Take speed of signal as 2×10^8 m/s.		3				
40	1	A data packet is sent over a network link of length 3000 km. The signal propagates at 2×10^8 meters per second. Calculate the propagation delay		3				
41	1	A 10 MB file needs to be transmitted over a network link with a transmission rate of 100 Mbps. Calculate the transmission delay for sending the entire file.		3				
42	1	A packet travels through three network links, each 500 km long, with a propagation speed of 2.5×10^8 m/s. If each link has a transmission rate of 50 Mbps and the packet size is 1 MB, calculate the total end-to-end delay (considering only transmission and propagation delays).		3				
43	1	A packet travels through four network nodes, where each node introduces a processing delay of 5 ms. The packet is sent over a link with a transmission rate of 10 Mbps and a propagation speed of 3×10^8 m/s. The link length is 1000 km, and the packet size is 2 MB. Calculate the total delay, including propagation, transmission, and processing delays		3				
44	1	A packet travels through three network links, each 5000 km long, with a propagation speed of 2.5×10^8 m/s. Each link has a transmission rate of 50 Mbps, and the packet size is 10 MB. Additionally, each network node introduces a processing delay of 20 ms and a queuing delay of 5 ms. Calculate the total end-to-end delay in seconds, considering transmission, propagation, processing, and queuing delays.						
45	1	A network link is 2500 km long, and the signal propagation speed is 2×10^8 m/s. If the link has a bandwidth of 1 Gbps, calculate: The propagation delay		3				
46	1	A network initially has a propagation delay of 20 ms for a link of 6000 km. If the link distance is increased to 9000 km with the same propagation speed, calculate the new propagation delay		3				
47	1	A data packet travels through a network with three different links: Link 1: 1000 km, transmission rate of 10 Mbps, propagation speed of 2×10^8 m/s Link 2: 2000 km, transmission rate of 50 Mbps, propagation speed of 2.5×10^8 m/s Link 3: 1500 km, transmission rate of 20 Mbps, propagation speed of 2×10^8 m/s If the packet size is 500 KB, calculate the total delay across all three links, considering both propagation and transmission delays		3				
48	1	A 5 MB file is segmented into packets of 1 KB each and transmitted over a network link with a transmission rate of 5 Mbps. Calculate the transmission delay for each packet and the total transmission delay to send the entire file		3				
49	1	A 500 KB file is sent over a 100 km link. The link has a transmission rate of 5 Mbps and a propagation speed of 2×10^8 m/s. Calculate the total delay (transmission + propagation delay).		3				
50	1	A video streaming server sends packets through a network with four consecutive links, each with its own characteristics: Link 1: 800 km, 20 Mbps transmission rate, 2×10^8 m/s propagation speed Link 2: 1000 km, 30 Mbps transmission rate, 2.5×10^8 m/s propagation speed Link 3: 600 km, 15 Mbps transmission rate, 2×10^8 m/s propagation speed Link 4: 1200 km, 40 Mbps transmission rate, 2.5×10^8 m/s propagation speed If the video packet size is 2 MB, calculate the total delay across all links, considering both propagation and transmission delays.		3				
51	1	A file of 8 MB is divided into packets of 512 KB each. If the transmission rate is 10 Mbps, calculate the total transmission time for sending all packets		3				
52	1	A data packet is sent over a fiber optic link of 2500 km with a propagation speed of 2×10^8 m/s. Calculate the propagation delay.		3				
53	1	A signal travels through a copper wire over a distance of 300 meters with a propagation speed of 2×10^8 m/s. Calculate the propagation delay.		3				
54	1	A data packet is sent over a network link of length 6500 km. The signal propagates at 2×10^8 meters per second. Calculate the propagation delay.		2				
55	1	List and describe the different types of networks (LAN, WAN, PAN, etc.). Provide examples of where each type might be used		3				
56	1	Explain the purpose of layered architecture in network design		3				
57	1	Write a short note on OSI Layer and its function		3				
58	1	Explain TCP/IP Layer		3				
59	1	Explain the functions of various network devices with neat diagram		3				
60	1	Compare Switch and Router		2				
61	1	Enlist function of Physical Layer		3				
62	1	Enlist function of Data link Layer		3				
63	1	Enlist function of Network Layer		3				
64	1	Enlist function of Transport Layer and Session Layer		3				
65	1	Enlist function of Application Layer and Presentation Layer		3				
66	1	Define Computer Network and its Components		3				
Unit-2 Physical Layer								
67	2	Which of the following represents the fastest data transmission speed?	A	1	Gbps	Kbps	Bps	Bandwidth
68	2	In a _____ connection , three or more devices share a link.	B	1	point to point	multipoint	(a) and (b)	None of the above.
69	2	Which of the following transmission directions listed is not a legitimate channel?	D	1	Simplex	Half Duplex	Full Duplex	Double Duplex
70	2	A _____ is a set of rules that governs data communication	C	1	forum	Standards	Protocol	None of the above.

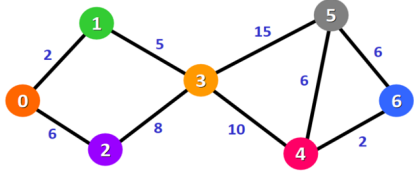
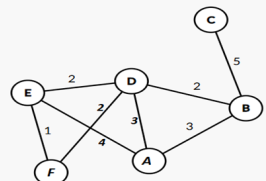
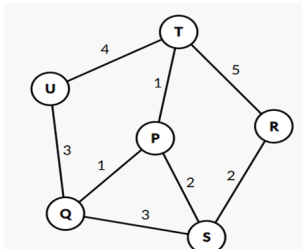
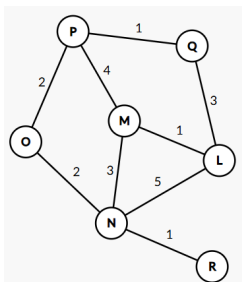
L.J Institute of Engineering and Technology, Ahmedabad. CN Practice book (SEM-V-2025 CE,IT,CSD,AI ML,AIDS,CS&IT,CSE,CST,CEA,RAI Engineering) Note : This Practic book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.								
Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
71	2	The method of communication in which transection take place in both directions, but only in one direction at a time,is called?	A	1	Half Duplex	Simplex	Full Duplex	Double Duplex
72	2	Network components are connected to the same cable in the _____ topology.	B	1	MESH	BUS	STAR	RING
73	2	Geometric arrangement of devices on the network is called	B	1	Protocols	Topology	Trailer	Lan
74	2	Which network topology requires a central controller or hub?	C	1	MESH	BUS	STAR	RING
75	2	_____ topology requires a multipoint connection.	B	1	MESH	BUS	STAR	RING
76	2	Which topology has the highest reliability?	A	1	MESH	BUS	STAR	RING
77	2	Frequency of failure and network recovery time after a failure are measures of the _____ of a network.	A	1	Reliability	Performance	Security	feasibility
78	2	The number of cables required for mesh topology is?	D	1	n	n-1	one backbone	n(n-1)/2
79	2	Suppose we want to create a network of 99 devices where there is a dedicated link between each and every device, then find how many links would be required?	A	1	4851	4951	5858	9999
80	2	What is the number of cable required for a bus topology?	D	1	n	n-1	n(n-1)/2	one backbone and n drop lines
81	2	A topology that is responsible for describing the geometric arrangement of components that make up the LAN.	C	1	complex	logical	physical	incremental
82	2	Signals with a frequency below 2MHz use _____ propagation.	A	1	ground	sky	line of sight	none of above
83	2	A parabolic dish antenna is a(n) _____ antenna	B	1	omnidirectional	unidirectional	bidirectional	horn
84	2	_____ cable consists of two insulated copper wires twisted together.	C	1	coaxial	Fiber-optic	Twisted-pair	None of the above
85	2	A(n) _____ medium provides a physcial conduit from one device to another.	A	1	guided	unguided	either(a) or (b)	None of the above
86	2	which cable consists of a central conductor and a shiled?	A	1	coaxial	Fiber-optic	Twisted-pair	None of the above
87	2	_____ cables carry data signals in the form of light.	B	1	coaxial	Fiber-optic	Twisted-pair	None of the above
88	2	Radio waves are _____	A	1	omnidirectional	unidirectional	bidirectional	both (a) & (b)
89	2	The physical layer translates logical communication requests from the _____ into hardware specific operations	A	1	Datalink	network	transport	application
90	2	In a fiber optics, the signal is _____ waves.	A	1	light	radio	Infrared	very low-frequency
91	2	The inner core _____ of an optical fiber is _____ in composition	C	1	copper	bimetalic	glass or plastic	liquid
92	2	Which transmission media _____ has the highest transmission speed in a network?	D	1	coaxial	electric cable	Twisted-pair	optical fiber
93	2	Which _____ of the following is not a transmission medium?	C	1	Telephone lines	Coaxial cables	modem	microwave systems
94	2	Which of the following is an advantage to using fiber optic data transmsstion?	D	1	Resistance to data theft	fast data transmission rate	low noice level	All of above
95	2	Which of the following communication lines is best suited to interactive processing application?	B	1	narrow band channel	Full duplex lines	simplex lines	Mixed Band channel
96	2	Guided media also known as _____	A	1	bound media	unguided media	unbound media	None of the above.
97	2	How many forms in twisted pair cable?	C	1	1	4	2	3
98	2	Which of the following cable doesn't come into guided media?	D	1	coaxial	Fiber-optic	Twisted-pair	microwave
99	2	which of the following are coaxial cable layers?	D	1	thinnet	thicknet	copper	A and B both
100	2	Which of the following connectors dosen't come into coaxial cable connectors?	B	1	barrel	UTP	BNC	T-connectors
101	2	What is the condition for total internal reflection ?	A	1	incidence angle is greater than critical angle	incidence angle is less than critical angle	incidence angle and critical angle equal	incidence angle is not related to critical angle
102	2	Which of the following tasks is performed by the Physical Layer	C	1	Logical addressing	Routing of packets	Transmission of raw bits over a communication channel	Error detection and correction
103	2	Which of the following transmission media is least affected by electromagnetic interference (EMI)?	B	1	Coaxial cable	Fiber optic cable	Twisted pair cable	Shielded twisted pair cable
104	2	Which of the following statements about microwave transmission is true?	D	1	Microwaves are used only for wired communication	Microwaves have frequencies below 1 MHz	Microwaves can easily pass through obstacles	Microwaves require line-of-sight propagation
105	2	In which network topology does each device have a dedicated point-to-point link to every other device?	A	1	MESH	BUS	STAR	RING
106	2	In a ring topology, data travels in:	B	1	Multiple paths simultaneously	One direction or sometimes two directions around the ring	Random directions	Star-shaped pattern
107	2	Define Topology.What are the common network topologies?		2				
108	2	Explain various topology and provide examples of where each might be useful.		2				
109	2	Describe the different types of transmission media used in networking		3				
110	2	What are the primary functions of the Physical layer?		3				
111	2	Assume 6 devices are arranged in a mesh topology. How many cables are needed? How many ports are needed for each device?		3				
112	2	Discuss the mode for propagating light along optical channels		3				
113	2	How are the guided media differing from unguided transmission media?		3				
114	2	Write a short note on Co-axial cable		3				
115	2	Write a short note on Fiber cable		3				
116	2	Write a short note on radio communication		3				
Unit-3 Data Link Layer and Logical Link Control Sublayer								
117	3	Which of the following tasks is not done by data link layer?	D	1	framing	error control	flow control	channel coding
118	3	When 2 or more bits in a data unit has been changed during the transmission, the error is called	B	1	random error	burst error	inverted error	double error
119	3	which of the following is not valid MAC address?	D	1	3A: 12: 34: B2: 11: 41	3A: 12: 3E: B2: 11: 41	3A: 12: 34: B2: 11: 4F	3A: 12: 34: B2: 1G: 41
120	3	Which MAC address is Broadcast MAC Address?	D	1	AA: BB :CC: 12: 14: 15	12: 23: 34: 45: 56:67	AA: BB: CC: DD: EE: FF	FF: FF: FF: FF: FF: FF
121	3	Which method is used for error detection as well as error correction?	C	1	LRC	VRC	Hamming code	CRC
122	3	In CRC the value of the key / Divisor is known by?	C	1	Sender	Receiver	Both	None
123	3	What is the size of receivers window for Go Back N ARQ?	B	1	2	1	N	None
124	3	The final code after encoding data bits 1101 into 7-bit even parity Hamming Code is	A	1	1010101	1111111	1000000	1100110
125	3	In bit stuffing when zero is stuffed?	B	1	After 4 ones	After 5 ones	After 3 Ones	After 6 Ones
126	3	Which Framing method is more efficient ?	B	1	Byte stuffing	Bit stuffing	Both	None
127	3	In _____ error correction, the receiver asks the sender to send the data again.	A	1	ARQ (Automatic Repeat Request)	FEC (Forward Error Correction)	Revert	None
128	3	The divisor in a cyclic code is normally called the _____	B	1	Degree	Generator	Redundency	None
129	3	What is the minimal Hamming distance between any two correct codewords?	C	1	1	2	3	4

L.J Institute of Engineering and Technology, Ahmedabad. CN Practice book (SEM-V-2025 CE,IT,CSD,AIML,AIDS,CS&IT,CSE,CST,CEA,RAI Engineering) Note : This Practic book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.								
Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
130	3	We add r redundant bits to each block to make the length $n = k + r$. the resulting n-bit blocks are called ?	C	1	Datawords	Blockwords	Codewords	None
131	3	What is the message length 'k' of a Hamming(7,4) code?	C	1	$2r-1$	$2r+r+1$	$2r-r-1$	$2r+1-r$
132	3	The _____ of errors is more difficult than the _____.	A	1	Correction; detection	detection;correcti on	creation;correcti on	creation;detection
133	3	Consider the data unit to be transmitted as 1001000111110000, divided into 4-bit segments. What is the checksum for this data unit? Ensure to add the carry if any.	A	1	0 1 0 1	0 0 0 0	0 0 1 1	0 1 0 0
134	3	if you wanted to transmit 17 data bits, how many parity bits are sufficient in hamming code?	A	1	5	4	3	2
135	3	What is the main advantage of CRC over VRC and LRC?	A	1	Higher error detection capability	Ability to detect and correct errors	Simplicity in implementation	Lower computational overhead
136	3	The stop and wait flow control method is the same as the sliding window method with a window size of _____.	1	1				
137	3	What will be the number of ACK packet sent by receiver if it recives 8th number packet.	9	1				
138	3	What is the size of receivers window for GB9 protocol if senders window size is 9.	1	1				
139	3	Sender has sent data 0100010001000011 but receiver has received 0101110101000011. find length of burst error.	5 bits	1				
140	3	Convert the polynomial $x^5 + x + 1$ into binary.	100011	1				
141	3	In CRC method the data sent by sender is 100100001. the size of original data is 6bits. Find size of the key.	4 bits	1				
142	3	In Stop and wait protocol every 4th packet is lost and we need to send total 10 packets so how many transmission it took to send all the packets?	13	2				
143	3	What is value of the checksum if the data to be sent 0101 0011 1000 is divided into three parts.	1110	3				
144	3	1. Given the data to be transmitted as 0100110 in Hamming code, Write the steps to find the 11-bit codeword that is sent to the receiver. Assume even parity. 2. If the codeword received is 01010010001, just mention the position of error, if any.		3				
145	3	Find value of LRC if the data to be sent is 11100111 11011101 00111001 10101001. consider size of one block 8 bits.	10101010	4				
146	3	Consider the following message M=1010001101. 1) The cyclic redundancy check for this message using the divisor polynomial $x^5 + x^4 + x^2 + 1$ is? 2) If the receiver receives 101000110101100, did it get the error free codeword? Just state Yes or No.		4				
147	3	What is value of the checksum if the data to be sent is 10011001111000100010010010000100. Consider size of each segment is 8bits.	11011010	5				
148	3	What is RTT time in Stop and Wait protocol if Propagation delay is 3.7 msec	7.4 msec	1				
149	3	In hamming code method suppose the size of the data bits is 7 then find the number of the redundant bits.	4bits	1				
150	3	The received data in even parity hamming code method is 10010100101. Identify the position of the error. Correct the data.	7th bit has error	4				
151	3	How Many ESC will be discarded if the received data has 4 ESC.	2ESC	1				
152	3	In stop and wait protocol what a sender will do if ack sent by receiver is lost?		1				
153	3	A bit stream 10011101 is transmitted using the standard CRC method. The generator polynomial is x^3+1 . What is the actual bit string transmitted?		3				
154	3	Code word received at the receiver side 100100001 and key is 1101 find if it has any error or not.		4				
155	3	In Byte stuffing how many ESC will be added if the data to be sent has ESC like data present?		2				
156	3	A bit stream 1101011011 is transmitted using the standard CRC method. The generator polynomial is x^4+x+1 . What is the actual bit string transmitted?		3				
157	3	Code word received at the receiver side - 100000001 and key is 1101 find if it has any error.		4				
158	3	How many number of MAC addresses are available?		1				
159	3	Encode the 4-bit data 1011 using (7,4) Hamming code. Show the positions of the parity bits and the final encoded message.		3				
160	3	A received Hamming codeword is 1010111. Identify whether there is an error in the codeword, and if so, determine the position of the error.		3				
161	3	Given a 7-bit Hamming codeword 1101101, calculate the values of the parity bits (P1, P2, P4, P8) and determine whether there are any errors in the codeword.		3				
162	3	In GB4, if every 6th packet being transmitted is lost and if we have to spend 10 packets then how many transmissions are required		3				
Unit-4 Data Link Layer and Medium Access Control Sublayer								
163	4	Why station need to wait random amount of time for re-transmission of the frame after collision in pure aloha?	B	1	To prevent other stations	To avoid more collisions	To detect other collisions	To get acknowledgment from the receiver
164	4	Which strategy is used by the pure aloha to prevent the congestion caused by continuous re-transmitted frames?	C	1	Aborts the re-transmission after 5 attempts	A station is allowed to re-transmit for 2 times	Restriction on the attempts for re-transmission	Re-transmission has no restriction
165	4	What is the vulnerable time of the pure aloha for fixed length of frames (T=frame time)?	A	1	$2*T$	$3*T$	$1/2*T$	$5*T$
166	4	The throughput for pure ALOHA is max when $G =$ _____.	A	1	0.5	1	2	none of above
167	4	Consider a 800 Kbps link. What is the maximum bandwidth (in Kbps) when pure aloha is used ?	B	1	294.4 Kbps	147.2 Kbps	194.4 Kbps	247.2 Kbps
168	4	The throughput for Pure ALOHA is _____.	C	1	$S = G \times e - G$	$S = 2G \times e - G$	$S = G \times e - 2G$	none of above
169	4	Maximum throughput for pure ALOHA is _____.	D	1	$S = G \times e - G$	0.368	$S = G \times e - 2G$	0.184
170	4	The throughput for slotted ALOHA is max when $G =$ _____.	A	1	1	1.5	0.5	none of above
171	4	The throughput for slotted ALOHA is _____.	A	1	$S = G \times e - G$	$S = 2G \times e - G$	$S = G \times e - 2G$	none of above
172	4	Maximum throughput for slotted ALOHA is _____.	D	1	$S = G \times e - G$	0.184	$S = G \times e - 2G$	0.368
173	4	Consider a 800 Kbps link. What is the maximum bandwidth (in Kbps) when slotted aloha is used ?	B	1	194.4 Kbps	294.4 Kbps	246.2 Kbps	146.2 Kbps
174	4	Radio broadcast on a common frequency to transmit uses _____ protocol.	D	1	Carrier Sense Multiple Access Protocols	Collision-Free Protocols	Limited-Contention Protocols	ALOHA Protocol

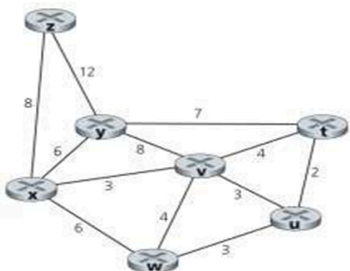
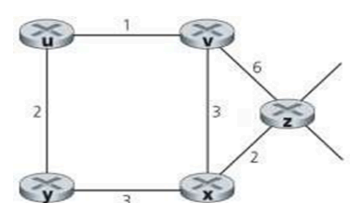
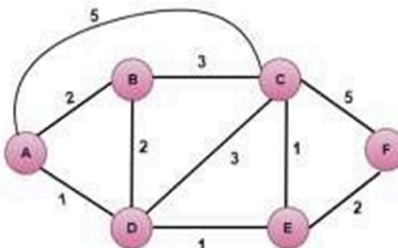
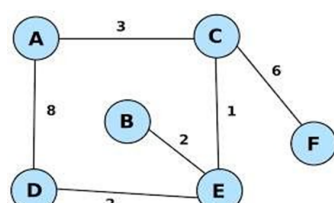
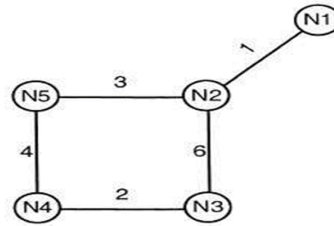
L.J Institute of Engineering and Technology, Ahmedabad. CN Practice book (SEM-V-2025 CE,IT,CSD,AI ML,AIDS,CS&IT,CSE,CST,CEA,RAI Engineering)								
Note : This Practic book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.								
Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
175	4	_____ is a medium access control protocol used for transmitting data via a shared network channel.	A	1	ALOHA Protocol	Framing	TCP	None of the above
176	4	What is the full form of CSMA?	D	1	Carrier Signal Multiple Access	Collision Sense Multiple Access	Channel Sense Multiple Access	Carrier Sense Multiple Access
177	4	Which of this is not a CSMA access mode?	C	1	1-Persistent	P-Persistent	A-Persistent	Non-Persistent
178	4	The CSMA/CA prevents collisions by:	D	1	Interframe Space (IFS)	Contention Window	Acknowledgement (ACK)	all of above
179	4	What is next step after sending jamming signal in CSMA/CD?	B	1	Wait for random time	Increment k	Wait for Tp	all of above
180	4	What is next step after sending frame in CSMA/CA?	A	1	Wait for timeout	Increment k	Wait for Tp	all of above
181	4	In CSMA/CD, after the fifth collision, what is the probability that a node chooses K = 4?	A	1	0.03125	0.0625	0.015625	none of above
182	4	Consider building a CSMA/CD network running at 1 Gbps over a 1 km cable with no repeaters. The signal speed in the cable is 200000km/sec. What is the minimum frame size?	B	1	1400 bytes	1250 bytes	1320 bytes	1211 bytes
183	4	In CSMA/CA the IFS can also be used to define priority of station or frame.	B	1	Contention Window	Inter Frame Space	Backoff Time	Frame Type
184	4	For wireless network _____ is not possible	C	1	static IP addressing	cabling infrastructure	collision detection	dedicated bandwidth
185	4	CSMA/CD can be used in _____ network	B	1	Wi-Fi	Ethernet	Bluetooth	Zigbee
186	4	What is start frame delimiter (SFD) in ethernet frame?	B	1	10101010	10101011	00000000	11111111
187	4	What IEEE standard is designated for Wi-Fi?	C	1	IEEE 802.1	IEEE 802.12	IEEE 802.11	IEEE 802.4
188	4	The maximum size of payload field in ethernet frame is _____	D	1	1000 bytes	1200 bytes	1300 bytes	1500 bytes
189	4	_____ is a standard communications protocol embedded in software and hardware devices, intended for building a local area network.	A	1	Ethernet	Wi-Fi	Bluetooth	Sneaker net
190	4	What IEEE standard is designated for Ethernet?	C	1	IEEE 802.1	IEEE 802.2	IEEE 802.3	IEEE 802.4
191	4	What is IEEE 802.3 primarily concerned with?	B	1	Wireless networking standards	Wired Ethernet networking standards	Bridging and switching technologies	Network security protocols
192	4	Which of the following is an Ethernet standard ?	A	1	10BASE-T	10BASE-P	10BASE-Q	10BASE-R
193	4	Convert the polynomial $x^5 + 2x + 1$ into binary	D	1	100001	101011	100011	None of the Above
194	4	If Frame contains data like ESC , Flag and ESC then after adding escape character the frame would look like	B	1	ESC,ESC,ESC,ESC ,ESC,ESC	ESC,ESC,ESC,flag, ESC,ESC	ESC,Flag,ESC,ESC ,ESC,ESC	ESC,ESC,flag,ESC, ESC,ESC
195	4	A bit-stuffing based framing protocol uses an 8-bit delimiter pattern of 01111110. If the output bit-string after stuffing is 011111100101, then the input bit-string is:	B	1	111110100	111110101	11111101	11111111
196	4	In CSMA/CD, after the sixth collision, how many distinct values can the random number R take?	D	1	1	4	15	64
197	4	In which layer of the OSI model does CSMA/CD operate?	A	1	Data Link Layer	Physical Layer	Session Layer	Presentation Layer
198	4	A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces 1000 frames per second?		2				
199	4	A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces 500 frames per second?		2				
200	4	A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces 250 frames per second?		2				
201	4	We have a pure ALOHA network with 100 stations. If $T_{fr}=1\mu s$, what is the number of frames each station can send to achieve the maximum efficiency.		2				
202	4	Consider a network using the pure ALOHA medium access control protocol, where each frame is of length 1000 bits. The channel transmission rate is 1 Mbps (10^6 bits per second). The aggregate number of transmissions across all the nodes (including new frame transmissions and retransmitted frames due to collisions) is modelled as a Poisson process with a rate of 1000 frames per second. Throughput is defined as the average number of frames successfully transmitted per second. The throughput of the network (rounded to the nearest integer) is		3				
203	4	If the frame transmission time is 20 ms, how many frames can be generated by the system in 1 second if $G=1$? For both Pure and Slotted Aloha.		3				
204	4	We have a slotted ALOHA network with 100 stations. If $T_{fr}=1\mu s$, what is the number of frames each station can send to achieve the maximum efficiency.		2				
205	4	A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200-kbps bandwidth. Find the throughput if the system (all stations together) produces 1000 frames per second.		2				
206	4	A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200-kbps bandwidth. Find the throughput if the system (all stations together) produces 500 frames per second.		2				
207	4	A slotted ALOHA network transmits 200-bit frames using a shared channel with a 200-kbps bandwidth. Find the throughput if the system (all stations together) produces 250 frames per second.		2				
208	4	The stations on a wireless ALOHA network are a maximum of 600 km apart. If we assume that signals propagate at 3×10^8 m/s, we find $T_p = (600 \times 10^3) / (3 \times 10^8) = 2$ ms. Now find the value of TB for K=1, K=2 and K=3		4				
209	4	A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the requirement to make this frame collision-free?		2				
210	4	A network using CSMA/CD has a bandwidth of 10 Mbps. If the maximum propagation time including the delays in the devices and ignoring the time needed to send a jamming signal is 25.6 μs , what is the minimum size of the frame?		3				
211	4	In csma/cd, after the 3rd collision, what is the probability that the value of k that a node chooses is 2?		2				
212	4	In CSMA/CD, after the 4th collision, what is the probability that a node chooses K=5? The result K=5 corresponds to a delay of how many seconds on Ethernet having $T_{fr} = 0.125$ microseconds?		2				
213	4	In csma/cd, after the sixth collision, what is the probability that a node chooses k=10? the result k=10 corresponds to a delay of how many seconds on Ethernet having $T_{fr} = 0.5$ microseconds?		2				

L.J Institute of Engineering and Technology, Ahmedabad. CN Practice book (SEM-V-2025 CE,IT,CSD,AI ML,AIDS,CS&IT,CSE,CST,CEA,RAI Engineering)								
Note : This Practic book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.								
Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
Unit-5 Introduction to Network Layer								
214	5	The network layer is concerned with _____ of data.	C	1	bits	frames	packets	bytes
215	5	Which one of the following is not a function of network layer?	D	1	routing	inter-networking	congestion control	error control
216	5	A 4 byte IP address consists of ?	C	1	only network address	only host address	network address & host address	network address & MAC address
217	5	IP is _____ protocol.	A	1	unreliable	connection-oriented	file transfer	secure communication
218	5	A telephone network is an example of a _____ network	B	1	Packet switched	Circuit switched	both of the mentioned	Message switching
219	5	Which of the following assertions is FALSE about the Internet Protocol (IP) ?	D	1	It is possible for a computer to have multiple IP addresses	IP packets from the same source to the same destination can take different routes in the network	IP ensures that a packet is discarded if it is unable to reach its destination within a given number of hops	The packet source cannot set the route of an outgoing packets; the route is determined only by the routing tables in the routers on the way
220	5	Computer networks that provide only a connection service at the network layer are called _____ while computer networks that provide only a connectionless service at the network layer are called _____	C	1	Broadcast, Multicast	Peer-to-peer, Client-server	Connection-oriented, Connectionless	None of the above
221	5	One of the header fields in an IP datagram is the Time to Live(TTL)field. Which of the following statements best explains the need for this field?	D	1	It can be used to prioritize packets	It can be used to reduce delays	It can be used to optimize throughput	It can be used to prevent packet looping
222	5	Which one of the following fields of an IP header is NOT modified by a typical IP router?	B	1	Checksum	Source Address	TTL	Length
223	5	An IPv4 packets has arrived with the first 8 bits as 01000010. Which of the following is correct?	D	1	The number of hops this packet can travel is 2	The total number of bytes in header is 16 bytes	The upper layer protocol is ICMP	The receiver rejects the packet
224	5	In an IPv4 packet, the value of HLEN is 1000 in binary. How many bytes of options are being carried by this packet?	B	1	4 bytes	12 bytes	16 bytes	8 bytes
225	5	In an IPv4 datagram, the M bit is 0, the value of HLEN is 10, the value of total length is 400 and the fragment offset value is 300. The position of the datagram, the sequence numbers of the first and the last bytes of the payload, respectively are	C	1	Last fragment, 2400 and 2789	First fragment, 2400 and 2759	Last fragment, 2400 and 2759	Middle fragment, 300 and 689
226	5	Which of the following fields in IPV4 datagram is not related to fragmentation?	A	1	Type of service	Fragment offset	Flags	Identification
227	5	Two computers C1 and C2 are configured as follows. C1 has IP address 203.197.2.53 and netmask 255.255.128.0. C2 has IP address 203.197.75.201 and netmask 255.255.192.0. which one of the following statements is true?	A	1	C1 assumes C2 is on same network, but C2 assumes C1 is on a different network	C1 and C2 both assume they are on the same network	C2 assumes C1 is on same network, but C1 assumes C2 is on a different network	C1 and C2 both assume they are on different networks.
228	5	If C1 and C2 both assume they are on different networks, which of the following actions is necessary for them to communicate?	D	1	Finding the IP address from the DNS	Finding the IP address of the default gateway	Finding the IP address that corresponds to a MAC address	Finding the MAC address that corresponds to an IP address
229	5	In virtual circuit network each packet contains	B	1	full source and destination address	a short VC number	both (a) and (b)	none of the mentioned
230	5	What is the 32-bit binary equivalent of the IP address 223.1.3.27?	A	1	11011111.00000001.00000011.000110011011	11011111.00000001.0000011.00011001	11011111.00000001.000011.00011111	11011111.00000001.000010.00011011
231	5	Which of the following is the broadcast address for a Class B network ID using the default subnetmask?	C	1	172.16.10.255	255.255.255.255	172.16.255.255	172.255.255.255
232	5	You have an IP address of 172.16.13.7 with a 255.255.255.128 subnet mask. What is your class of address, subnet address, and broadcast address?	B	1	Class A, Subnet 172.16.13.0, Broadcast address 172.16.13.127	Class B, Subnet 172.16.13.0, Broadcast address 172.16.13.127	Class B, Subnet 172.16.13.0, Broadcast address 172.16.13.255	Class B, Subnet 172.16.0.0, Broadcast address 172.16.255.255
233	5	If you wanted to have 12 subnets with a Class C network ID, which subnet mask would you use?	D	1	255.255.255.252	255.255.255.255	255.255.255.248	255.255.255.240
234	5	If you wanted to have 10 subnets with a Class C network ID, which subnet mask would you use?	D	1	255.255.255.252	255.255.255.255	255.255.255.248	255.255.255.240
235	5	Given the following IP address and network mask, what IS the broadcast address? IP : 160.168.30.100 Net Mask : 255.255.240.0	A	1	160.168.31.255	160.168.30.255	160.168.24.03	160.168.25.53
236	5	The forwarding table of a router is shown below.A packet addressed to a destination address 200.150.68.118 arrives at the router. It will be forwarded to the interface with ID _____.	C	1	1	2	3	4
237	5	Suppose datagrams are limited to 1,500 bytes (including header) between source Host A and destination Host B. Assuming a 20-byte IP header and a 20-byte TCP header, how many datagrams would be required to send an MP3 consisting of 4 million bytes.		2				
238	5	IP Packets on a certain network can carry a maximum of only 500 bytes in the data portion. An application using TCP/IP on a node on this network generates a TCP segment with 1,000 bytes in the data portion. How many IP packets are transmitted to carry this TCP segment, and what are their sizes (including the header)?		2				
239	5	An IP datagram of size 1000 bytes arrives at a router. The router has to forward this packet on a link whose MTU (maximum transmission unit) is 100 bytes. Assume that the size of the IP header is 20 bytes. The number of fragments that the IP datagram will be divided into for transmission is?		4				
240	5	Consider sending a 2400-byte datagram into a link that has an MTU of 700 bytes. Suppose the original datagram is stamped with the identification number 422. How many fragments are generated? What are the values in the various fields in the IP datagram(s) generated related to fragmentation?		4				
241	5	An IP router with a Maximum Transmission Unit (MTU) of 200 bytes has received an IP packet of size 520 bytes with an IP header of length 20 bytes. The values of the relevant fields in the IP header.		3				

L.J Institute of Engineering and Technology, Ahmedabad. CN Practice book (SEM-V-2025 CE,IT,CSD,AI ML,AIDS,CS&IT,CSE,CST,CEA,RAI Engineering)								
Note : This Practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.								
Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
242	5	Consider an IP packet with a length of 4,500 bytes that includes a 20-byte IPv4 header and a 60-byte TCP header. The packet is forwarded to an IPv4 router that supports a Maximum Transmission Unit (MTU) of 600 bytes. Assume that the length of the IP header in all the outgoing fragments of this packet is 20 bytes. Assume that the fragmentation offset value stored in the first fragment is 0. The fragmentation offset value stored in the third fragment is _____.		3				
243	5	A 3200-byte datagram is to be sent over a link with an MTU of 900 bytes. The datagram has the identification number 512. How many fragments are generated? What are the values in the various fields in the IP datagram(s) generated related to fragmentation?		4				
244	5	Host A is connected to a router R1. R1 is connected to another router R2 and R2 is connected to host B.  Suppose that a TCP message that contains 1000 bytes of data including 20 bytes of TCP header is passed to the IP layer at the host A for delivery to B. Maximum frame size between link A-R1 is 1500 bytes including a 15-byte frame header. Link R1-R2 can support a maximum frame size 700 bytes including 10-byte frame header and link R2 B can support maximum frame size of 600 bytes including a 12-byte frame header. find the total length, DF, MF and fragment offset fields of the IP header in the last packet transmitted over the link R1-R2? [Assume 20 B of IP header]		3				
245	5	Host A sends a UDP datagram containing 8880 bytes of user data to host B over an Ethernet LAN. Ethernet frames may carry data up to 1500 bytes (i.e. MTU = 1500 bytes). Size of UDP header is 8 bytes and size of IP header is 20 bytes. There is no option field in IP header. How many total number of IP fragments will be transmitted and what will be the contents of offset field in the last fragment?		2				
246	5	Suppose users share a 3 Mbps link. Also, suppose each user requires 150 kbps when transmitting, but each user transmits only 10 percent of the time. a) When circuit switching is used, how many users can be supported? b) For the remainder of this problem, suppose packet switching is used. Find the probability that a given user is transmitting.		3				
247	5	An ATM network uses a token bucket scheme for traffic shaping. A new token is put into the bucket every 5 μsec. Each token is good for one cell, which contains 48 bytes of data. What is the maximum sustainable data rate?		2				
248	5	Consider a router that interconnects three subnets: Subnet 1, Subnet 2, and Subnet 3. Suppose all of the interfaces in each of these three subnets are required to have the prefix 223.1.17/24. Also suppose that Subnet 1 is required to support at least 60 interfaces, Subnet 2 is to support at least 90 interfaces, and Subnet 3 is to support at least 12 interfaces. Provide three network addresses (of the form a.b.c.d/x) that satisfy these constraints.		3				
249	5	A large number of consecutive IP addresses are available starting at 198.16.0.0. Suppose that four organizations, A, B, C, and D, request 4000, 2000, 4000, and 8000 addresses, respectively, and in that order. For each of these, give the first IP address assigned, the last IP address assigned, and the mask in the w.x.y.z/s notation.		4				
250	5	IPv6 uses 16-byte addresses. If a block of 1 million addresses is allocated every picosecond, how long will the addresses last?		4				
251	5	Find the class of each address. a. 000000001 00001011 00001011 11101111 b. 11000001 10000011 00011011 11111111 c. 14.23.120.8 d. 252.5.15.111		2				
252	5	Given the IP address 193.1.2.0 with a subnet mask of 255.255.255.128: a) What is the subnet ID of Subnet-1? b) What is the direct broadcast address of Subnet-1? c) Calculate the total number of usable hosts in Subnet-1 and Subnet-2, justify the answer. d) For Subnet-2, if you needed to create two additional subnets within it while maximizing the number of hosts in each new subnet, what would be the new subnet masks for these subnets? Provide the subnet IDs and broadcast addresses for both new subnets.		4				
253	5	An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows: a. The first group has 64 customers; each needs 256 addresses. b. The second group has 128 customers; each needs 128 addresses. c. The third group has 128 customers; each needs 64 addresses. Design the sub blocks and find out how many addresses are still available after these allocations		4				
254	5	A network with the IP address 10.1.1.0/22 needs to support at least 100 hosts. Is this subnet sufficient? If not, what subnet mask should be used to accommodate at least 100 hosts? Justify your answer.		3				
255	5	You have been allocated a network which needs to be divided into three subnets having hosts A) 189 B) 1589 C) 3998. also find network and broadcast address for each subnets. How many addresses are wasted in each subnets. The ip address 11.12.13.14/8 is one ip address of given network.		4				
256	5	You have been assigned the IP address 10.0.0.0/8. If you need to create 16 subnets, what subnet mask should you use?		2				
257	5	An organization is granted the block 190.76.0.0/16. The administrator wants to create 1024 subnets using 10 bits. What is the first and last addresses in last subnet which can assign to host ?		3				
258	5	Consider a IP address 201.24.58.69 in classful address, if the number of 1's in directed broadcast address is a and number of 1's in network ID, of the given IP address is b, value of a + b		3				
259	5	In the network 143.128.67.235/20, if x represent the decimal value of 3rd octet and y represent the decimal value of 4th octet of last IP address assigned to any host, then value of x * y is		3				
260	5	In a class B subnet, we know the IP address of one host and the mask as given below: IP address = 125.134.112.66 Mask = 255.255.224.0 What is the first address (Network address)?		2				

L.J Institute of Engineering and Technology, Ahmedabad. CN Practice book (SEM-V-2025 CE,IT,CSD,AIML,AIDS,CS&IT,CSE,CST,CEA,RAI Engineering) Note : This Practic book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.								
Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
261	5	An IP datagram of length (including a header of 20 bytes) 1800 bytes needs to cross an Ethernet followed by a WAN to reach its destination. The MTU for Ethernet is 1500 bytes. For the WAN, the MTU is given to be 576 bytes. The more-fragment value, fragment offset value and total length field value stored in all the fragments generated are?		5				
Unit-6 Routing in Network Layer								
262	6	Count-to-Infinity problem occurs in _____.	A	1	Distance vector Routing	Link state	Dijkstra's Algorithm	none of the above
263	6	Which one of the following routing algorithm can be used for network layer design?	D	1	Dijkstra's Algorithm	distance vector routing	link state routing	all of the mentioned
264	6	Routing tables of a router keeps track of _____.	D	1	MAC Address Assignments	Port Assignments to network devices	Distribute IP address to network devices	Routes to use for forwarding data to its destination
265	6	_____ command is used to manipulate TCP/IP routing table.	A	1	route	Ipconfig	Ifconfig	Traceroute
266	6	In _____ forwarding, the full IP address of a destination is given in the routing table.	C	1	next-hop	network-specific	host-specific	default
267	6	The idea of address aggregation was designed to alleviate the increase in routing table entries when using _____.	B	1	classful addressing	classless addressing	both a and b	none of the above
268	6	The routing processor of a router performs the _____ layer functions of the router.	B	1	physical and data link	network	transport	none of the above
269	6	In 223.1.1.0/24,the /24 notations, is known as a _____.	C	1	subnet mask	CIDR	both a and b	none of the above
270	6	In Multicast Routing Protocol,_____ is used to broadcasts packets, but creates loops in the systems.	B	1	Forwarding	Flooding	Backwarding	none of the above
271	6	What is the natural mask for a class C Network?	B	1	255.255.255.1	255.255.255.0	255.255.255.254	255.255.255.255
272	6	In multicast routing, each involved router needs to construct a _____ path tree for each group.	C	1	average	longest	shortest	none of the above
273	6	The Open Shortest Path First (OSPF) protocol is an intradomain routing protocol based on _____ routing.	B	1	distance vector	link state	path vector	none of the above
274	6	In distance vector routing, the updating packets are sent _____.	A	1	periodically	when there is a change	either a or b	neither a nor b
275	6	Which algorithm is used by the router in the link state routing to find the shortest paths?	C	1	OSPF	RIP	Dijkstra algorithm	BGP
276	6	In link state routing, the updating packet conveys the knowledge of the router about _____.	B	1	the whole internetwork	the neighborhood	either a or b	neither a nor b
277	6	Generate the shortest path from node 0 to all the other nodes in the graph using Dijkstra algorithm. 		5				
278	6	You are running Dijkstra's algorithm on the edge-weighted digraph below, starting from vertex A. Find shortest path to all other nodes from A. 		5				
279	6	Find a shortest path using Distance Vector routing from node Q. 		4				
280	6	Find a shortest path using Bellman-Ford algorithm from node N. 		5				

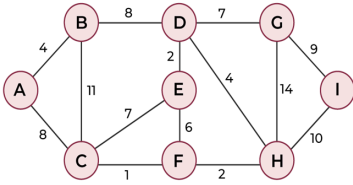
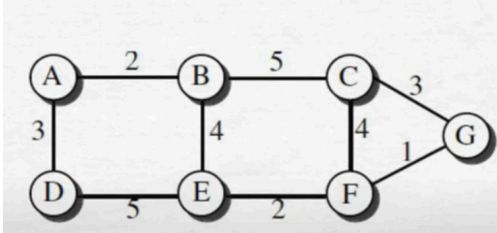
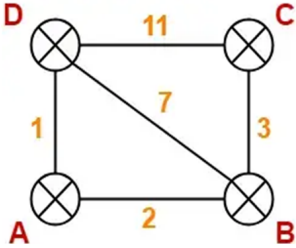
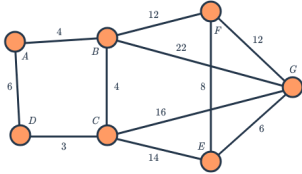
Note : This Practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.

Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
281	6	Consider the following network. With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from x to all network nodes. 		5				
282	6	Consider the network shown below, and assume that each node initially knows the costs to each of its neighbors. Consider the distance-vector algorithm and show the distance table entries at node z. 		5				
283	6	Find a shortest path using Link State Routing. 		5				
284	6	Find a shortest path using Dijkstra's shortest- path algorithm. 		3				
285	6	Consider a network with five nodes, N1 to N5, as shown below The network uses a Distance Vector Routing Protocol. once the Route have stabilized, the distance vectors at different nodes are as following N1: (0, 1, 7, 8, 4) N2: (1, 0, 6, 7, 3) N3: (7, 6, 0, 2, 6) N4: (8, 7, 2, 0, 4) N5: (4, 3, 6, 4, 0) Each distance vector is the distance of best known path at that instance to nodes, N1 to N5, where the distance to itself is 0. Also, all links are symmetric and the cost is identical in both directions. In each round, all nodes exchange their distance vectors with their respective neighbors. Then all nodes update the distance vectors. In between two rounds, any change in cost of a link will cause the two incident nodes to change only that entry in their distance vectors. The cost of link N2-N3 reduces to 2 (in both directions). After the next round updates, what will be the new distance vector at node, N3? 		5				

Note : This Practice book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.

Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
286	6	Consider a network with 6 routers R1 to R6 connected with links having weights as shown in the following diagram. All the routers use the distance vector based routing algorithm to update their routing tables. Each router starts with its routing table initialized to contain an entry for each neighbour with the weight of the respective connecting link. After all the routing tables stabilize, how many links in the network will never be used for carrying any data?		4				
287	6	Find the shortest path in the following diagram from A to D.		4				
288	6	Apply distance vector algorithm in the following diagram		4				
289	6	Apply distance vector algorithm in the given diagram		4				
290	6	Apply distance vector algorithm in the following diagram.		4				
291	6	Suppose that forwarding tables are all established and then C-E link fails. What update will be there in the C and E tables after failure and which tables information will be exchanged? what will be the table of A after C and E have reported the news?		5				

Note : This Practic book is only for reference purpose. LjU Test question paper may not be completely set from this practice book.

Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
292	6	Take a graph and find the shortest path from the source to all nodes using distance vector algorithm. 		5				
293	6	Find a shortest path using link state routing. 		4				
294	6	Find the shortest path from node C to remaining nodes using distance vector routing algorithm. 		4				
295	6	Consider the following network. With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from A to all network nodes. 		5				

Unit-7 Introduction to Transport Layer

296	7	Transport layer aggregates data from different applications into a single stream before passing it to	A	1	Network Layer	Datalink Layer	Application Layer	Physical Layer
297	7	Transport layer protocols deals with	B	1	application to application communication	process to process communication	node to node communication	man to man communication
298	7	A _____ is a TCP name for a transport service access point.	A	1	port	pipe	node	socket
299	7	What are the functions of transport layer?	D	1	Multiplexing/De multiplexing	Connectionless and Connection Oriented Services	Congestion Control	all of the mentioned
300	7	UDP stand s for _____.	A	1	User datagram protocol	Union data Protocol	Unity device protocol	Username data protocol
301	7	Port adress range is _____ in transport layer	B	1	0 to 61,536	0 to 65,535	0 to 62,535	0 to 65,353
302	7	Consider the following statements. I. TCP connections are full duplex. II. TCP has no option for selective acknowledgment III. TCP connections are message streams.	A	1	Only I is correct	Only I and II are correct	Only II and III are correct	All of I, II and III are correct
303	7	Which of the following is not a field in TCP header?	D	1	Sequence number	Window size	Checksum	Fragment offset
304	7	User datagram protocol is called connectionless because _____.	A	1	all UDP packets are treated independently by transport layer	it sends data as a stream of related packets	it is received in the same order as sent order	it sends data very quickly
305	7	Transport layer is responsible for process to process delivery using _____.	C	1	IP Address	MAC Address	Port Address	none of the above
306	7	Transport layer can identify the symptoms of overload nodes using	B	1	Traffic control	Flow control	Byte orientation	Data Integrity
307	7	With regard to Internet security, the fullform of SSL is _____ and it provides security at _____ layer.	C	1	Simple Socket Layer, Transport	Secure Socket Layer, Application	Secure Socket Layer, Transport	Secure Socket Layer, Application

L.J Institute of Engineering and Technology, Ahmedabad. CN Practice book (SEM-V-2025 CE,IT,CSD,AI ML,AIDS,CS&IT,CSE,CST,CEA,RAI Engineering)								
Note : This Practic book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.								
Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
308	7	The transport layer may be responsible for flow and error control, like the	B	1	Physical layer	Data link layer	Subnet layer	Application layer
309	7	A communication device that combines transmissions from several I/O devices into one line is a	C	1	Concentrator	Modifier	Multiplexer	Full duplex file
310	7	Which of the following are transport layer protocols used in networking?	C	1	TCP and FTP	UDP and HTTP	TCP and UDP	HTTP and FTP
311	7	The sharing of a medium and its link by two or more devices is called?	B	1	Fully duplexing	Multiplexing	Micropleixng	Duplexing
312	7	If there are n signal sources of same data rate, then the TDM link has slots?	A	1	n	n/2	n*2	2^n
313	7	If link transmits 4000 frames per second, and each slot has 8 bits, the transmission rate of circuit using TDM is?	A	1	32kbps	500bps	500kbps	32bps
314	7	_____ does not provide reliable end to end communication.	B	1	TCP	UDP	Both TCP and UDP	Neither TCP nor UDP
315	7	ARQ stands for _____	A	1	Automatic Repeat Request	Automatic Request Repeat	Application Repeat Request	Application Request Repeat
316	7	What is the efficiency of a system using synchronous TDM if 2 of the 5 devices have nothing to send?	C	1	0.2	0.5	0.6	1
317	7	The ports ranging from 0 to 1023 are called _____ ports.The ports ranging from 1024 to 49151 are called _____ ports.The ports ranging from 49152 to 65535 are called _____ ports	C	1	well-known,registered, static or public	unknown,registered, dynamic or private	well-known,registered, dynamic	well-known,unregistered, dynamic or private
318	7	What is the minimum and maximum size of TCP Header in bytes?	B	1	40, 60	20, 60	20, 30	60, 120
319	7	Explain the duties of transport layer.		3				
320	7	Alyssa P. Hacker and Ben Bitdiddle communicate over a link with capacity of 100 pkts / sec. The latency (RTT) on this link is 100 ms. If a sliding window protocol with acknowledgement packets is used, and there is a FIXED window size of 4 packets, what is the maximum rate of traffic on the link?		7				
321	7	What are the two types of protocols used in transport layer?		2				
322	7	Enlist services provided by transport layer. What is socket? Explain its importance at transport layer protocol		7				
Unit-8 Transport Layer Protocols								
323	8	Let the size of congestion window of a TCP connection be 32 KB when a timeout occurs. The round trip time of the connection is 100 msec and the maximum segment size used is 2 KB. The time taken (in msec) by the TCP connection to get back to 32 KB congestion window is	A	2	1100 to 1300	800 to 1000	1400 to 1600	None of the mentioned
324	8	On a TCP connection, current congestion window size is Congestion Window = 4 KB. The window size advertised by the receiver is Advertise Window = 6 KB. The last byte sent by the sender is LastByteSent = 10240 and the last byte acknowledged by the receiver is LastByteAcked = 8192. The current window size at the sender is	B	2	2048 bytes	4096 bytes	6144 bytes	None of the mentioned
325	8	Assume that the bandwidth for a TCP connection is 1048560 bits/sec. Let 'a' be the value of RTT in milliseconds (round off to nearest integer) after which the TCP window scale option is needed. Let 'b' be the maximum possible window size with window scale option. Then the values of 'a' and 'b' are	C	2	63 ms, 65535 x 2^14	63 ms, 65535 x 2^16	500 ms, 65535 x 2^14	None of the mentioned
326	8	The main difference between TCP and UDP is	A	1	UDP is a datagram where as TCP is a connection oriented service	UDP is connection oriented where as TCP is datagram service	TCP is an Internet protocol where as UDP is an ATM protocol	TCP and UDP both are connectionless service protocol
327	8	End-to-end connectivity is provided from host-to-host in:	D	1	Session layer	Network layer	Data link layer	Transport layer
328	8	Security based connection is provided by which layer?	A	1	Transport layer	Application layer	Network layer	Session layer
329	8	In transport layer, message is divided into transmittable	A	1	Segments	Packets	Frames	Data
330	8	Which of the following statement is incorrect?	C	1	The data link layer handles the transfer of data between the ends of a physical link	The presentation layer transforms information from machine format to the understandable format	The transport layer provides for any format, translation or code conversion necessary to put the data into an intelligible format	All of the above
331	8	In TCP, an ACK segment, if carrying no data, consumes Sequence number(s).	A	1	no	one	two	three
332	8	TCP and UDP are called	C	1	Application Protocols	Session Protocols	Transport Protocols	Network Protocols
333	8	In TCP, SYN segment consumes Sequence number(s).	B	1	no	one	two	three
334	8	A client that wishes to connect to an open server tells its TCP that it needs to be connected to that particular server. The process is called	A	1	Active open	Active close	Passive close	Passive open
335	8	Transmission Control Protocol (TCP), needs buffers for	A	1	Storage	Request	Connections	Transmission
336	8	UDP packets have a fixed-size header of bytes.	B	1	16	8	40	32
337	8	The Source port number on the UDP user datagram header defines	C	1	The sender computer	The receiver computer	The process running on the sending computer	The process running on receiving computer
338	8	Transmission Control Protocol (TCP), A connection-oriented transport protocol establishes a	B	1	Virtual Communication	Virtual Path	Virtual Flow	Virtual Flow
339	8	In User Datagram Protocol (UDP), the user datagrams are not	C	1	Channeled	Transformed	Numbered	Addressed
340	8	Beyond IP, UDP provides additional services such as	D	1	Routing and switching	Sending and receiving of packets	Multiplexing and demultiplexing	Demultiplexing and error checking
341	8	“Total length” field in UDP packet header is the length of	D	1	Only UDP header	Only data	Only checksum	UDP header plus data
342	8	Suppose a TCP connection is transferring a file of 1000 bytes. The first byte is numbered 10001. What is the sequence number of the segment if all data is sent in only one segment?	B	1	10000	10001	12001	11001
343	8	Connection establishment in TCP is done by which mechanism?	B	1	Flow control	Three-Way Handshaking	Forwarding	Synchronization
344	8	What allows TCP to detect lost segments and in turn recover from that loss?	D	1	Sequence number	Acknowledgment number	Checksum	Both Sequence & Acknowledgment number
345	8	Beyond IP, UDP provides additional services such as	D	1	Routing and switching	Sending and receiving of packets	Multiplexing and demultiplexing	Demultiplexing and error checking

L.J Institute of Engineering and Technology, Ahmedabad. CN Practice book (SEM-V-2025 CE,IT,CSD,AI ML,AIDS,CS&IT,CSE,CST,CEA,RAI Engineering)								
Note : This Practic book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.								
Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
346	8	The port number is “ephemeral port number”, if the source host is	D	1	NTP	Echo	Server	Client
347	8	UDP is protocol.	A	1	Transport Layer Connectionless	Session Layer Connectionless	Transport Layer Connection Oriented	Session Layer Connection Oriented
348	8	Consider the following statements regarding the OSI model: (1) Application Layer provides the control structure for communication between applications; establishes , manages and terminates connection between cooperating applications. (2) Datalink Layer provides for the reliable transfer of information across the physical link. (3) Transport Layer provides end-to-end error recovery and flow control of data. Which of the above statements are correct?	D	2	1 and 2 only	1 and 3 only	1, 2 and 3 only	2 and 3 only
349	8	Consider a long-lived TCP session with an end-to-end bandwidth of 1 Gbps. The session starts with a sequence number of 1234. The minimum time before this sequence number can be used again	B	1	30-32	34 - 35	20-23	40-45
350	8	The transport layer protocols use for real time multimedia, file transfer are	C	1	UDP, UDP	TCP, TCP	UDP, TCP	TCP, UDP
351	8	The transport layer protocols use for DNS and email respectively are	C	1	UDP, UDP	TCP, TCP	UDP, TCP	TCP, UDP
352	8	Consider the following statements regarding the slow start phase of the TCP congestion control algorithm. Note that cwnd stands for the TCP congestion window and MSS denotes the maximum segment size. (i) The cwnd increases by 2 MSS on every successful acknowledgement (ii) The cwnd increases doubles on every successful acknowledgement (iii) The cwnd increases by 1 MSS on every round trip time (iv) The cwnd approximately doubles on every round trip time. Which one of following is correct?	C	2	Only (ii) and (iii) are true	Only (i) and (iii) are true	Only (iv) is true	Only (i) and (iv) are true
353	8	What is the value of the receiver window (rwnd) for host A if the receiver, host B, has a buffer size of 7000 bytes and 1000 bytes of received and unprocessed data in bytes?	C	1	1000	7000	6000	2000
354	8	What is the size of the window in bytes for host A if the value of rwnd is 3000 bytes and the value of cwnd is 5500 bytes?	D	1	1000	7000	6000	3000
355	8	The following is a dump of a UDP header in hexadecimal format. CB84002C 001C0AEC What is the destination port number?	D	2	43	32	42	44
356	8	The following is a dump of a UDP header in hexadecimal format. DB84002C001E0A1C What is the length of the data?	C	2	19	18	22	20
357	8	Compare connectionless services and connection oriented services.		4				
358	8	Explain factors measuring the network traffic performance.		3				
359	8	Let the size of congestion window of a TCP connection be 32KB when a timeout occurs. The round trip time of the connection is 100 msec and the maximum segment size used is 2KB. The time taken (in msec) by the TCP connection to get back to 32KB congestion window is	1100-1300	5				
360	8	How TCP protocol provides reliability?		3				
361	8	Explain the error control mechanism in TCP?		4				
362	8	List the services of end to end services.		3				
363	8	What are the advantages of using UDP over TCP?		4				
364	8	Which protocol uses both TCP and UDP? and How?		3				
365	8	What is the value of the receiver window (rwnd) for host A if the receiver, host B, has a buffer size of 6000 bytes and 1000 bytes of received and unprocessed data?	5000	3				
366	8	What is the value of the receiver window (rwnd) for host A if the receiver, host B, has a buffer size of 5000 bytes and 1000 bytes of received and unprocessed data?	4000	3				
367	8	What is the size of the window for host A if the value of rwnd is 3000 bytes and the value of cwnd is 4500 bytes?	3000 bytes	3				
368	8	What is the size of the window for host A if the value of rwnd is 4000 bytes and the value of cwnd is 4500 bytes?	4000 bytes	3				
369	8	What is the size of the window for host A if the value of rwnd is 3000 bytes and the value of cwnd is 3500 bytes?	3000 bytes	3				
370	8	Flow Control versus Congestion control		4				
371	8	The following is a dump of a UDP header in hexadecimal format. AA8400D001A001A i. What is the source port number? ii. What is the destination port number? iii. What is the total length of the user datagram? iv. What is the length of the data?		4				
372	8	The following is a dump of a UDP header in hexadecimal format. ABAB00D001C001C i. What is the source port number? ii. What is the destination port number? iii. What is the total length of the user datagram? iv. What is the length of the data?		4				
373	8	The following is a dump of a UDP header in hexadecimal format. CCDD00D001C001C i. What is the source port number? ii. What is the destination port number? iii. What is the total length of the user datagram? iv. What is the length of the data?		4				
374	8	The following is a dump of a UDP header in hexadecimal format. CB8400D001C001C i. What is the source port number? ii. What is the destination port number? iii. What is the total length of the user datagram? iv. What is the length of the data?		3				
375	8	Explain the TCP Segment structure and justify the importance of its field values.		7				
376	8	What are various reliable data transfer mechanisms and for what purpose are they used?		4				
377	8	How many packets overhead while doing the data communication using TCP? Draw the TCP connection establishment and termination process with diagram		7				
378	8	Explain how ip address will get by host A if it wants to connect with		3				

L.J Institute of Engineering and Technology, Ahmedabad. CN Practice book (SEM-V-2025 CE,IT,CSD,AI ML,AIDS,CS&IT,CSE,CST,CEA,RAI Engineering)								
Note : This Practic book is only for reference purpose. LJU Test question paper may not be completely set from this practice book.								
Sr No	Unit Number	Question_Text	MCQ Answer	Marks	Option A	Option B	Option C	Option D
Unit-9 Application Layer								
379	9	Address used at application layer is	A	1	URL	port address	MAC address	IP address
380	9	The packet of information at the application layer is called	B	1	Packet	Message	Segment	Frame
381	9	Which one of the following is an architecture paradigms?	D	1	Peer to peer	Client-server	HTTP	Both Peer-to-Peer & Client-Server
382	9	Application developer has permission to decide the following on transport layer side	C	1	Transport layer protocol	Maximum buffer size	Both Transport layer protocol and Maximum buffer size	None of the mentioned
383	9	Application layer offers service.	A	1	End to end	Process to process	Both End to end and Process to process	None of the mentioned
384	9	E-mail is	C	1	Loss-tolerant application	Bandwidth-sensitive application	Elastic application	None of the mentioned
385	9	Which is a time-sensitive service?	D	1	File transfer	File download	E-mail	Internet telephony
386	9	Electronic mail uses which Application layer protocol?	A	1	SMTP	HTTP	FTP	SIP
387	9	The translates internet domain and host names to IP address.	A	1	domain name system	routing information protocol	network time protocol	internet relay chat
388	9	Which one of the following allows a user at one site to establish a connection to another site and then pass keystrokes from local host to remote host?	C	1	HTTP	FTP	Telnet	TCP
389	9	In one of the pairs of protocols given below, both the protocols can use multiple TCP connections between the same client and the server. Which one is that?	A	1	HTTP, FTP	HTTP, TELNET	FTP, SMTP	HTTP, SMTP
390	9	Which of the following protocol is used for transferring electronic mail messages from one machine to another?	D	1	TELNET	FTP	SNMP	SMTP
391	9	A web client sends a request to a web server. The web server transmits a program to that client ans is executed at client. It creates a web document. What are such web documents called?	A	1	Active	Static	Dynamic	Passive
392	9	Pick the odd one out.	D	1	File transfer	File download	E-mail	Interactive games
393	9	The translates internet domain and host names to IP address.	A	1	domain name system	routing information protocol	internet relay chat	network time protocol
394	9	Application layer protocol defines:	C	1	types of messages exchanged	message format, syntax and semantics	rules for when and how processes send	all of the mentioned
395	9	Which is not a application layer protocol?	C	1	HTTP	SMTP	UDP	HTTPS
396	9	Which is not a application layer protocol?	D	1	HTTP	SMTP	FTP	TCP
397	9	What are the responsibilities of application layer?		3				
398	9	What are the steps to transfer a mail message?		3				
399	9	List out common TCP/IP protocols.		3				
400	9	What is HTTP and what port does it use?		3				
401	9	Explain the functions of SMTP.		4				
402	9	What is the difference between a user agent (UA) and a mail transfer agent (MTA)?		3				
403	9	What is the Domain name system responsible for?		3				
404	9	What are the four main properties of HTTP?		4				
405	9	What are the responsibilities of Application Layer?		4				
406	9	What are the types of messages in HTTP transaction?		4				
407	9	Explain the functions of SMTP.		3				
408	9	Explain role of application layer while sending an e-mail?		3				
409	9	List five different examples involving role of application layer.		4				
Unit-10 Network Designing and Monitoring								
410	10	If you want to find the number of routers between a source and destination, the utility to be used is	D	1	route	Ipconfig	Ifconfig	Traceroute
411	10	command is used to manipulate TCP/IP routing table.	A	1	route	Ipconfig	Ifconfig	Traceroute
412	10	Ping sweep is a part of	B	1	route	N-map	Ifconfig	Traceroute
413	10	Wireshark is a tool.	A	1	network protocol analysis	network connection security	connection analysis	defending malicious packet-filtering
414	10	Wireshark filter for filter by IP	A	1	ip.addr == 10.10.50.1	ip.dest == 10.10.50.1	ip.src == 10.10.50.1	None of mentioned
415	10	Wireshark filter for filter by destination Ip	B	1	ip.addr == 10.10.50.1	ip.dest == 10.10.50.1	ip.src == 10.10.50.1	None of mentioned
416	10	Wireshark filter for filter by Sorce IP	C	1	ip.addr == 10.10.50.1	ip.dest == 10.10.50.1	ip.src == 10.10.50.1	None of mentioned
417	10	Wireshark filter for filter by Port	A	1	tcp.port == 25	tcp.port.port==25	tcp.dstport == 25	None of mentioned
418	10	Wireshark filter for filter by URL	B	1	tcp.port == 25	http.host == "host name"	tcp.dstport.port == 25	None of mentioned
419	10	Wireshark filter for filter by Destination Port	C	1	tcp.port == 25	tcp.port.port==25	tcp.dstport == 25	None of mentioned
420	10	What will be the filter if I want to find out packet from or for specific ip address 8.8.8.8	A	1	ip.addr==8.8.8.8	ipaddress=8.8.8.8	ip address=8.8.8.8	ip „addr==8.8.8.8